Experiement-18

18. DES Subkey Generation Using Two 28-Bit Halves

Aim:

To write a C program to demonstrate DES key scheduling using two 28-bit halves and the prescribed shift schedule.

Procedure:

1. Read the two 28-bit key halves, C and D.
2. Initialize the DES left-shift schedule.
3. Rotate both halves according to each round's shift count.
4. Display the two halves after every round.
5. Use PC-2 to select 48 bits when implementing complete DES subkey generation.

Code (q18.c)

```
#include <stdio.h>

int main(void) {
    int C[28], D[28];
    int s[16] = {1,1,2,2,2,2,2,2,
                 1,2,2,2,2,2,2,1};
    int i, r, j, t;

    printf("Enter 28 bits for C: ");
    for (i = 0; i < 28; i++)
        if (scanf("%1d", &C[i]) != 1 || C[i] > 1) return 1;

    printf("Enter 28 bits for D: ");
    for (i = 0; i < 28; i++)
        if (scanf("%1d", &D[i]) != 1 || D[i] > 1) return 1;

    for (r = 0; r < 16; r++) {
        for (j = 0; j < s[r]; j++) {
            t = C[0];
            for (i = 0; i < 27; i++) C[i] = C[i+1];
            C[27] = t;
```

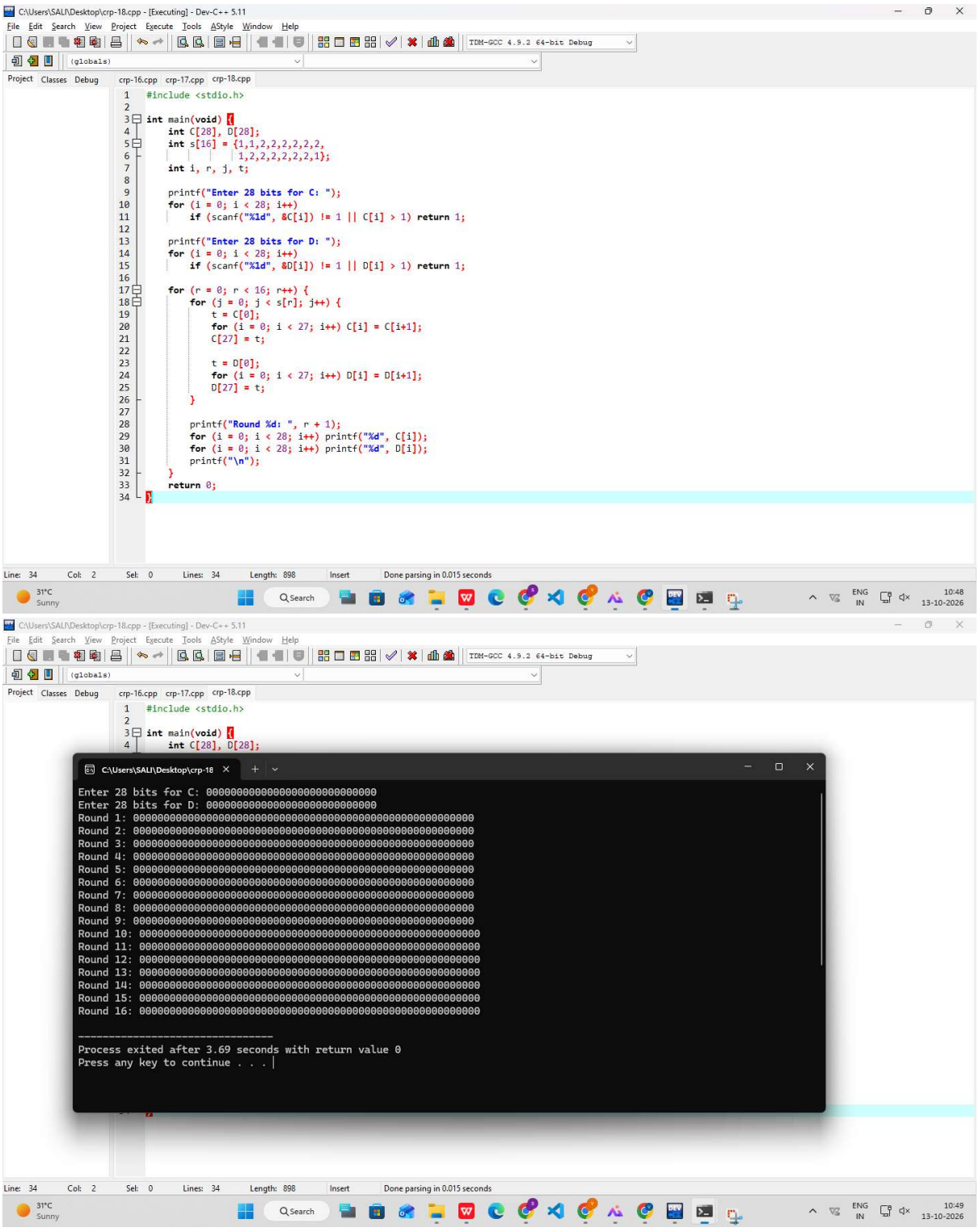
```
            t = D[0];
            for (i = 0; i < 27; i++) D[i] = D[i+1];
            D[27] = t;
        }

        printf("Round %d: ", r + 1);
        for (i = 0; i < 28; i++) printf("%d", C[i]);
        for (i = 0; i < 28; i++) printf("%d", D[i]);
        printf("\n");
    }
    return 0;
}
```

Sample Input

```
Enter 28 bits for C: 0000000000000000000000000000
Enter 28 bits for D: 0000000000000000000000000000
```

Sample Output



Result:

The two 28-bit halves are rotated using the DES shift schedule. PC-1 and PC-2 are required to generate complete DES subkeys from an original key.